

Pavel Mosharev

Physicist and algorithm researcher | Quantum and quantum-inspired optimization

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Profile

Current work studies quantum and quantum-inspired algorithms for discrete optimization through mechanism-focused analysis and reproducible numerical experiments. Background spans theoretical physics, university teaching, and industrial algorithm R&D at Huawei.

Professional Experience

2025–Present	Postdoctoral Researcher University of Science and Technology of China, School of Physical Sciences (Hefei; based in Shenzhen, China) - Study quantum and quantum-inspired discrete optimization through Ising/QUBO models, variational quantum algorithms, energy landscapes, and controlled numerical experiments. - Develop collaborative research with quantum-science groups in Shenzhen; contribute to experiment design, validation, manuscripts, and student research advising.	USTC
Sep 2025–Mar 2026	Visiting Scholar Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen, China; six-month concurrent appointment.	GBA Quantum Science Center
2023–2025	Postdoctoral Research Fellow Huawei Technologies Co., Ltd., 2012 Laboratories, Shenzhen, China - Researched quantum and quantum-inspired optimization through solver simulation and benchmarking, heuristic analysis, Ising-model construction and preprocessing, and related SAT/MILP investigations. - Developed protected-routing algorithms for network resilience as the main applied line; also contributed to wireless optimization. Two algorithms were transferred to a Huawei product line; received the Future Star Award 2024.	Huawei
2022–2023	Algorithm Engineer Huawei Russian Research Institute, Moscow, Russia Developed quantum-inspired constrained-optimization methods and protected-routing algorithms for optical/IP networks; received the Huawei Spark Award 2022.	Huawei
2020	Independent Contractor / Research Project Gazprom Neft PJSC, Moscow, Russia Delivered Python tools to assess CCE/PVT laboratory data and estimate reservoir-fluid saturation pressure.	Gazprom Neft
2016–2022	Assistant, later Senior Lecturer National Research University “Moscow Power Engineering Institute” (MPEI), Moscow, Russia Taught university mathematics, including analysis, linear algebra, differential equations, and probability.	MPEI
Feb–Jun 2021	Senior Lecturer Gubkin University, Moscow, Russia; taught general physics and supervised laboratory practice.	Gubkin University
Sep–Dec 2021	Instructor HSE University, Moscow, Russia; led undergraduate Linear Algebra seminars for economics students.	HSE University

Education

2016–2021	PhD in Physics (Candidate of Sciences) Lomonosov Moscow State University, Moscow, Russia	MSU
2010–2016	Specialist in Physics, with honours Lomonosov Moscow State University, Moscow, Russia	MSU

Technical Skills

- Algorithms and optimization:** Discrete and combinatorial optimization; graph algorithms; constrained and multi-objective optimization; solver benchmarking.
- Quantum and quantum-inspired methods:** Variational quantum algorithms/QAOA; Ising/QUBO/HUBO modeling and reduction; simulated bifurcation and quantum-annealing-inspired algorithms; circuit simulation.
- Scientific computing:** Python for numerical modeling, data analysis, visualization, and validation; Qiskit and MindQuantum; C++ (selected projects).
- Research engineering:** Reproducible experiments, testing and provenance; Git; Linux/WSL and remote computing workflows; AI-assisted coding and research workflows with coding agents.

Languages

- Russian — Native
- English — Advanced (IELTS Academic 7.5, 2024)
- Chinese — Upper-intermediate (HSK 5, 2026)

Selected Research and Technical Contributions

- **Quantum optimization and algorithm behavior:** Work combines theoretical analysis with exact simulations and energy-landscape studies to understand how shallow-circuit and related algorithms explore discrete problems.
- **Industrial graph optimization:** Developed cost-driven pruning methods for constrained protected routing and led their later reproducible evaluation on public network benchmarks.
- **Higher-order Ising modeling:** Co-conceived a reduction method for general-order Ising models and co-developed its code and numerical experiments; a minimal public reference implementation accompanies the work.
- **Applied scientific computing:** Built Python methods for CCE/PVT data processing, quality checks, and saturation-pressure estimation in a Gazprom Neft research project.

Selected Publications

1. X. Zhu, Z. Zou, F. Jin, **P. Mosharev**, et al. "Combinatorial optimization enhanced by shallow quantum circuits with 104 superconducting qubits." *National Science Review*, 13(9), nwag124, 2026. Equal contribution.
2. **P. A. Mosharev**, C.-M. Lee, X. Shu, X. Zhang, and M.-H. Yung. "Cost-Driven Prunings for Iterative Solving of Constrained Routing Problem With SRLG-Disjoint Protection." *IEEE Transactions on Networking*, 34, 92–106, 2026.
3. C. Mao, **P. Mosharev**, Y. Wang, and M.-H. Yung. "A reduction scheme for general-order Ising-like Hamiltonians in quantum heuristic solvers." *Quantum Information Processing*, 25(10), 319, 2026.
4. G. Kalachev, **P. Mosharev**, Z. Zou, P. Panteleev, and M.-H. Yung. "Pilot-Wave Simulator: Exact Classical Sampling from Ideal and Noisy Quantum Circuits up to Hundreds of Qubits." *Quantum*, 10, 2173, 2026. Corresponding author.
5. X.-Z. Tao, **P. Mosharev**, and M.-H. Yung. "Multi-Objective Optimization by Quantum-Annealing-Inspired Algorithms." arXiv:2604.26477, 2026. Preprint; manuscript submitted; equal contribution.
6. Q.-G. Zeng, X.-P. Cui, B. Liu, Y. Wang, **P. Mosharev**, and M.-H. Yung. "Performance of quantum annealing inspired algorithms for combinatorial optimization problems." *Communications Physics*, 7, 249, 2024.
7. A. A. Lobanov, **P. A. Mosharev**, et al. "Mathematical analysis of the behavior of PV isotherms of reservoir fluids." *Geology, Geophysics and Development of Oil and Gas Fields*, 8(380), 39–50, 2023 (in Russian; title translated).
8. O. V. Kechkin and **P. A. Mosharev**. "A General Harmonic Solution in Dilaton Electrodynamics: An Exact Expression for the Fields and the Generalized Lorentz Force." *Moscow University Physics Bulletin*, 75(5), 427–433, 2020.